

DevOps- Assignment-4

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Create a local git repository. Create an index.html file in this repository. Create a Dockerfile to build a docker image using this index.html and httpd web server docker image. Commit the code.

Create a repository on the GitHub. Push the code from the local repository to GitHub repository.

Go to your system. Clone the repository. Build the image. Push it to docker hub. Create a deployment and service using this image.

Create a Jenkinsfile in the local git directory to create a pipeline job on the Jenkins server. The Jenkins server will first fetch the new code from the GitHub repository. Then build the docker image. Then it will push the image to docker hub. Then it will update the image in the minikube deployment.

1. Creating Local repo, adding index.html and dockerfile, and pushing to github repo

```
PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops
$ mkdir devops-assignment5

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops
$ git init
Initialized empty Git repository in D:/ai-devops/.git/

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops (master)
$ notepad index.html

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops (master)
$ mkdir devops-assignment4

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops
$ cd devops-assignment4

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4
$ git init
Initialized empty Git repository in D:/ai-devops/devops-assignment4/.git/

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ notepad index.html

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ cat index.html
This is version 1 for DevOps Assignment 4
PRN: 8033
PRN: 8038
PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ |
```

```

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ notepad Dockerfile

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ mv Dockerfile.txt Dockerfile

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ cat Dockerfile
FROM httpd
COPY index.html /usr/local/apache2/htdocs/index.html
PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ git add .

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ git commit -m "index.html v1 dockerfile added"
[master 9939f64] index.html v1 dockerfile added
1 file changed, 2 insertions(+)
create mode 100644 Dockerfile

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ git remote add origin https://github.com/maurya-Priti/devops-assignment4.git

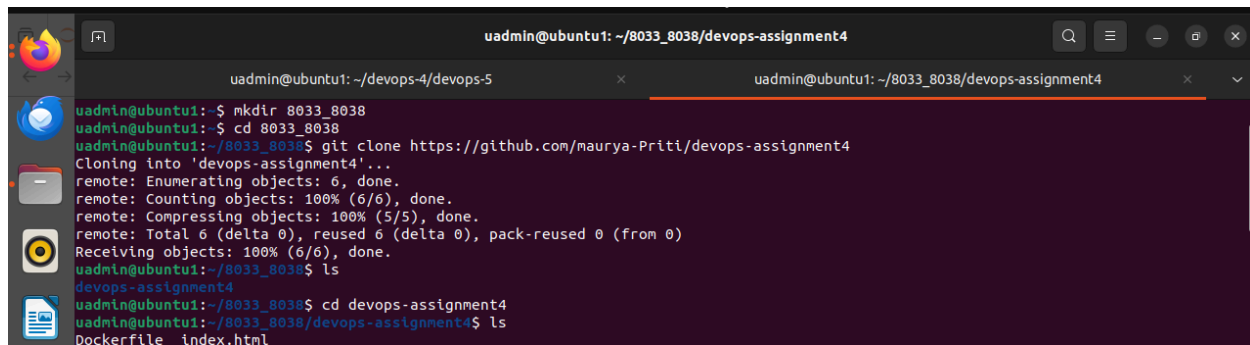
PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (master)
$ git branch -M main

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (main)
$ git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 596 bytes | 298.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/maurya-Priti/devops-assignment4.git
* [new branch]      main -> main

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (main)
$

```

2. Creating clone of the remote repo in ubuntu system



```

uadmin@ubuntu1: ~/8033_8038/devops-assignment4
uadmin@ubuntu1: ~/devops-4/devops-5
uadmin@ubuntu1: ~/8033_8038$ mkdir 8033_8038
uadmin@ubuntu1: ~/8033_8038$ cd 8033_8038
uadmin@ubuntu1: ~/8033_8038$ git clone https://github.com/maurya-Priti/devops-assignment4
Cloning into 'devops-assignment4'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
uadmin@ubuntu1: ~/8033_8038$ ls
devops-assignment4
uadmin@ubuntu1: ~/8033_8038$ cd devops-assignment4
uadmin@ubuntu1: ~/8033_8038/devops-assignment4$ ls
Dockerfile  index.html

```

3. Creating docker image and pushing to docker hub

```
uadmin@ubuntu1: ~/8033_8038/devops-assignment4
uadmin@ubuntu1: ~/devops-4/devops-5
uadmin@ubuntu1: ~/8033_8038/devops-assignment4$ sudo docker build -t pritiimaurya/devops-assignment4:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 59.9kB
Step 1/2 : FROM httpd
--> 393435ee1a31
Step 2/2 : COPY index.html /usr/local/apache2/htdocs/index.html
--> 6e8842fdb34b
Successfully built 6e8842fdb34b
Successfully tagged pritiimaurya/devops-assignment4:v1
uadmin@ubuntu1: ~/8033_8038/devops-assignment4$ docker push pritiimaurya/devops-assignment4:v1
The push refers to repository [docker.io/pritiimaurya/devops-assignment4]
tag does not exist: pritiimaurya/devops-assignment4:v1
uadmin@ubuntu1: ~/8033_8038/devops-assignment4$ docker push pritiimaurya/devops-assignment4:v1
The push refers to repository [docker.io/pritiimaurya/devops-assignment4]
02986002441a: Pushed
```

4. Create deployment and service, displaying website

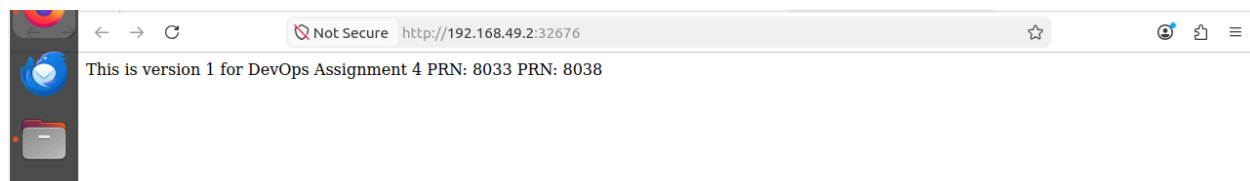
```
uadmin@ubuntu1: ~/8033_8038/devops-assignment4$ kubectl create deployment devops-assignment-app1 --image=pritiimaurya/devops-assignment4:v1 --port=8080
deployment.apps/devops-assignment-app1 created

uadmin@ubuntu1: ~/8033_8038/devops-assignment4$ kubectl expose deployment devops-assignment-app1 --type=NodePort --port=8080 --target-port=80
service/devops-assignment-app1 exposed
uadmin@ubuntu1: ~/8033_8038/devops-assignment4$ minikube service devops-assignment-app1
```

NAMESPACE	NAME	TARGET PORT	URL
default	devops-assignment-app1	8080	http://192.168.49.2:32676

Opening service default/devops-assignment-app1 in default browser...

```
uadmin@ubuntu1: ~/8033_8038/devops-assignment4$
```



5. Adding jenkinsfile to github repo from local repo

```
PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (main)
$ notepad Jenkinsfile

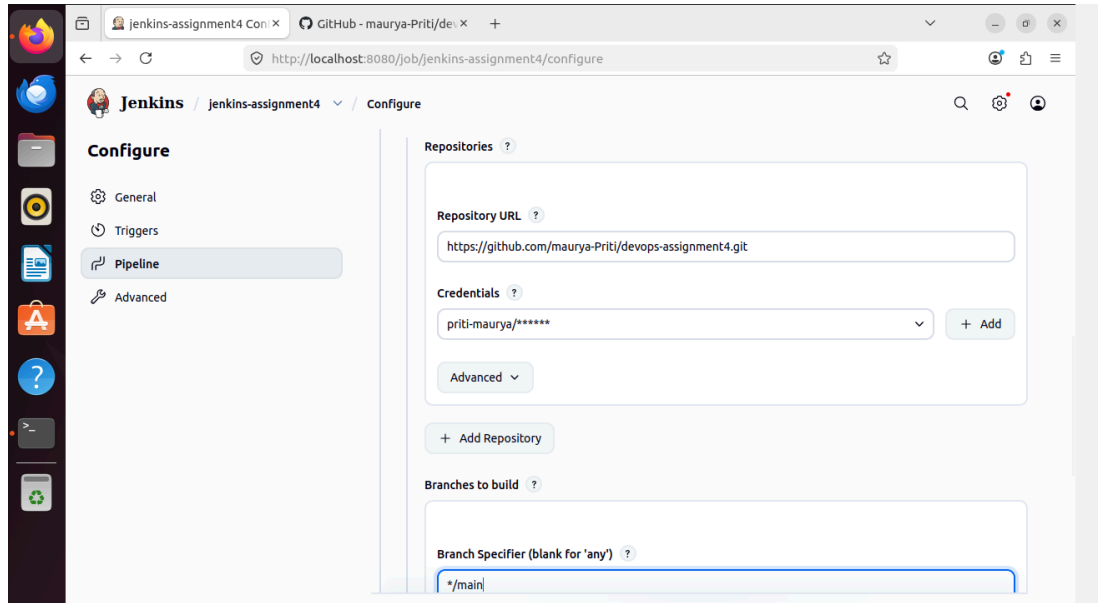
PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (main)
$ git add .

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (main)
$ git commit -m "jenkinsfile added"
[main fe08c6c] jenkinsfile added
1 file changed, 41 insertions(+)
create mode 100644 Jenkinsfile.txt

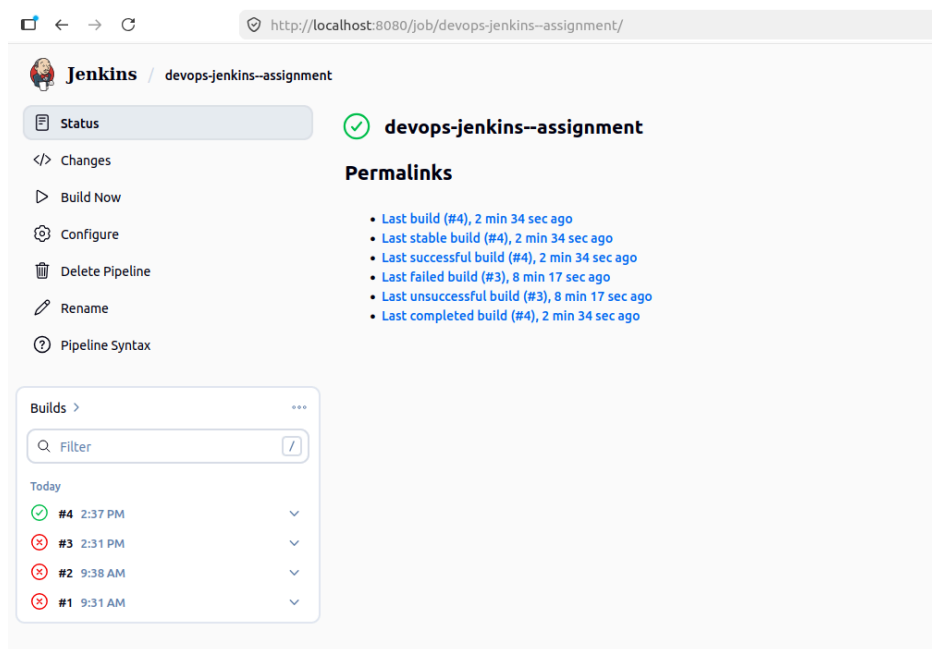
PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 802 bytes | 802.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/maurya-Priti/devops-assignment4.git
9939f64..fe08c6c main -> main

PGCP-AI@PGCP-AI13 MINGW64 /d/ai-devops/devops-assignment4 (main)
$ |
```

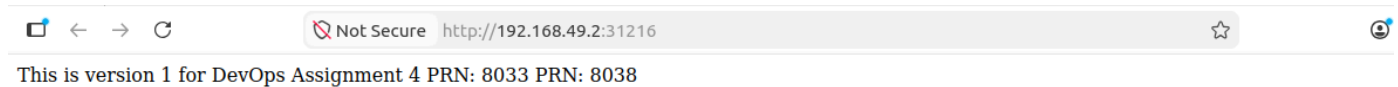
6. Creating jenkins pipeline



Building the pipeline



Website



7. Modifying the index.html file and committing , then building the pipeline

The screenshot shows the Jenkins web interface for build #5 of the 'devops-jenkins--assignment' job. The console output is displayed in a light blue box with a dark blue border. The output shows the pipeline starting with a checkout of the Jenkinsfile from a GitHub repository. It then proceeds to a stage with a declarative checkout of the SCM. The pipeline then runs a series of commands to set up the git environment, including resolving the git directory, setting the remote origin, and configuring the git tool. The pipeline then fetches changes from the remote repository and updates the deployment image. The final output shows the deployment image updated and the pipeline finished successfully.

```
Started by user maurya-priti
Obtained Jenkinsfile from git https://github.com/maurya-Priti/devops-assignment4.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/devops-jenkins--assignment
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential priti123
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/devops-jenkins--assignment/.git #
timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/maurya-Priti/devops-assignment4.git # timeout=10
Fetching upstream changes from https://github.com/maurya-Priti/devops-assignment4.git
> git --version # timeout=10
> git --version # 'git version 2.34.1'
using GIT_ASKPASS to set credentials
48565bdf4e87: Pushed
5: digest: sha256:fe39dc1e5f5e350e39b88d04d44dce6018b8264ff9cd09698e8ba02e7a863c53 size: 1830
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (update-minikube)
[Pipeline] sh
+ kubectl set image deployment/devops-assignment-app1 devops-assignment4=priti123/devops-assignment4:5
deployment.apps/devops-assignment-app1 image updated
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.555.3

Build successful, updated version

The screenshot shows a web browser window with the URL 'http://192.168.49.2:31216'. The page content is as follows:

This is version 2 for **DevOps Assignment 4**

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